Geo-Insight

Overlookedness as Disagreement:
A Multi-Attribute, Multi-Stakeholder Gap-Ranking System for Humanitarian Crises

Datathon 2026

April 2026

Abstract

In 2026, agencies are asking \$33B to reach 135 M people; the 2025 appeal raised \$12B, the lowest in a decade, with 19 contexts below 50 % coverage. When nearly every response is underfunded, a single coverage ratio stops discriminating. We recast overlookedness as a joint distribution: a coverage-gap score computed under four donor-preference profiles, plus an intra-crisis equity term (sector-coverage Gini) and a cross-stakeholder disagreement term (rank IQR across profiles). The resulting four-cell typology separates *consensus-overlooked*, *contested*, and *sector-starved* crises that single-number methods conflate. Scoring is *evidence-bounded*: missing signals contribute zero, so incomplete data cannot inflate rank. The method is implemented over a typed semantic layer of 71 properties across five derivation levels. Validated against two independently-curated benchmarks: 100 % top-5 overlap with OCHA's CERF Underfunded Emergencies on the strict pool ($n = 18$), 80 % on the widest pool ($n = 57$).

1 Motivation

The 2026 Global Humanitarian Overview asks \$33B for 135 M people, with a hyper-prioritised \$23B floor targeting 87 M lives.[1] The 2025 appeal raised \$12B. Coverage ratios have collapsed — DRC ∼22 %, Yemen and Somalia ∼24 %, Sudan from 69 % (2024) to 35 % (2025). Nineteen contexts are flagged as severely underfunded, up from eight in 2021. A single coverage ratio cannot distinguish between 19 countries that are all below 50 % coverage; the useful variation is no longer *whether* a crisis is underfunded but *how*, and *for whom*.

Three kinds of overlookedness become distinguishable once scoring moves from scalar to joint distribution: *forgotten* (low funding, low attention — Angola drought 2023), *contested* (high attention, persistent decline — Sudan 2025), and *sector-starved* (adequate aggregate, one sector at 20 %). Each warrants a different response — multi-year pooled funding, targeted advocacy, sector-level reallocation — and current tools conflate them.

2 Related Work

Rye & Aktas [1] propose EDIS, a MADM/AHP/MAUT framework whose Table 13 demonstrates rank divergence under different decision-maker preferences but stops short of aggregating it into a meta-metric. Three papers converge on equity as the open gap. Abdulrashid et al. [2] (2026 review, §7.3) list equity-aware prioritisation as one of only five open research items. Sahebjamnia et al. [3] call for *"egalitarian and utilitarian measures"*. Vargas Florez et al. [4] give the equation template (Eqs. 9–12) for fair-distribution penalties; we generalise from inter-regional to inter-cluster inequality within a single country response. Griffith et al. [5] argue for interpretable

[1] https://www.unocha.org/publications/report/world/global-humanitarian-overview-2026-enesfr

analytics in time-constrained humanitarian operations, which we take as a hard constraint: every added term must decompose cleanly and be explainable to a non-technical analyst. No paper in the review aggregates per-decision-maker rank divergence into an operational disagreement meta-score, and none operationalises intra-crisis equity as a Gini on cluster coverage ratios within a single country response. Both are small, defensible extensions.

3 Data

Six upstream datasets, with declared provenance. Five from OCHA: HNO (people in need per country and sector), HRP (plan metadata), FTS (requirements, funding received, incoming flows per donor), CoD-PS (population), and CBPF (pooled-fund allocations and donor contributions). One from the EU Joint Research Centre: INFORM Severity, as 67 monthly global snapshots consolidated into long-format CSVs (109 countries, 1–5 ordinal severity plus sub-indicators: affected, displaced, fatalities, phase-level PIN splits, access-limited/restricted populations). Two externally-curated lists for validation only, not training: CERF Underfunded Emergencies (22 country selections across 2024–25) and CARE *Breaking the Silence* (annual top-10 by media attention, not funding). Countries missing more than half of the six scoring signals are withheld rather than imputed.

4 Methodology

Unit of analysis is country-year. A country-year is eligible if it has either positive FTS requirements or a non-null 24-month INFORM severity baseline; it is *scored* if completeness $c(u) \geq \frac{1}{2}$.

Six-attribute vector per country-year:

- a_1 coverage shortfall $= 1 - \min(\text{fund}/\text{req}, 1)$
- a_2 gap per PIN $= \max(0, \text{req} - \text{fund})/\text{PIN}$
- a_3 need intensity $= \text{PIN}/\text{population}$
- a_4 severity: INFORM 1–5 ordinal
- a_5 donor concentration (HHI over FTS single-destination incoming flows)
- a_6 intra-crisis equity: PIN-weighted Gini over sector coverage ratios,

$$a_6(u) = \frac{\sum_{i,j} n_i n_j |k_i - k_j|}{2\,(\sum_c n_c)^2\,\bar{k}_w}, \quad \bar{k}_w = \frac{\sum_c n_c k_c}{\sum_c n_c}.$$

Each attribute is min–max normalised within the pool; utility U_i is identity for a_1–a_5 and mildly concave on a_6.

Four donor-preference profiles represent plausible stakeholder weightings (Table 1).

Profile	a_1	a_2	a_3	a_4	a_5	a_6
CERF	0.15	0.15	0.20	0.30	0.10	0.10
ECHO	0.15	0.20	0.15	0.15	0.10	0.25
USAID	0.15	0.15	0.15	0.15	0.25	0.15
NGO	0.10	0.15	0.15	0.15	0.10	0.35

Table 1: Profile weight vectors w_p.

Evidence-bounded gap score. For crisis u under profile p,

$$G_p(u) \;=\; \sum_{i \in \mathcal{O}(u)} w_{p,i}\, U_i(\tilde{a}_i(u)), \tag{1}$$

where $\mathcal{O}(u)$ is the set of signals actually observed. We sum rather than renormalise by $\sum_{\mathcal{O}} w$: a country's maximum possible score is bounded by its observed weight mass, so incomplete data cannot inflate rank. The cost — systematic under-ranking of poorly-documented crises — is declared via the completeness flag $c(u) = |\mathcal{O}(u)|/6$ attached to every score. The alternative (renormalising) collapses the CERF-UFE top-5 overlap from 100 % to 0 % on our pool; see §6.

Disagreement band and four-cell typology. The four profile-specific ranks yield a median rank $\widetilde{\text{rank}}(u)$ and an IQR across profiles. A median split on IQR against a median split on a_6 produces the four-cell typology (Table 2).

	Donors agree	**Donors disagree**
Sectors uneven	consensus sector-starved	contested sector-starved
Sectors even	consensus overlooked	contested balanced

Table 2: Four-cell typology from rank-IQR × cluster Gini.

Temporal decomposition. Chronic overlookedness is a multi-year rolling mean of the gap; acute is the deviation of the current year from that mean. Both are registered as Level-4 properties (§5).

5 Typed Semantic Layer

Every analytic quantity is a named property registered in a single specification file. Seventy-one properties at five levels: **L1 Observations** (22 raw primitives), **L2 Ratios** (18 scale-normalised signals), **L3 Inequality** (10 distribution indices: HHI, Gini, concentration measures), **L4 Trends** (12 windowed aggregates: baselines, deltas, trend slopes, persistence), **L5 Composites** (9: four profile-specific gap scores, balanced, median rank, rank-IQR, INFORM composite, completeness). Each property declares its formula, source dataset, input lineage, unit, granularity, and known failure modes. No analytic column exists in code without a registered declaration; a registry class reads the specification at import time so every displayed value carries its provenance. Eight lenses group subsets of the 71 properties by analytic question; six view modes render each lens (ranked atlas, PCA, clustering, per-country profile, cross-lens radar, validation). All 48 lens-by-mode cells render without per-cell code.

6 Validation

Two human-curated external benchmarks, neither used as training data. CERF UFE (22 countries across 2024–25, 17 in 2025 alone) maps to the underfunding-and-consensus axis. CARE BTS (annual top-10) maps partially, by design, to the forgotten/sector-starved axis — high-attention crises are by construction absent from CARE's list, so rank correlation is expected to drift as the pool grows.

Pool	**CERF 2024–25 (22)**	**CERF 2025 (17)**	**CARE BTS (10)**
Strict, $c = 1.0$ (18)	Top-5 **100 %** · @10 80 %	Top-5 **100 %** · @10 80 %	@10 10 %
Extended, $c \geq 2/3$ (33)	Top-5 80 % · @10 80 %	Top-5 80 % · @10 70 %	@10 20 %
All, $c \geq 1/2$ (57)	Top-5 **80 %** · @10 **80 %**	Top-5 80 % · @10 70 %	@10 20 %

Table 3: Agreement with external benchmarks across three evidence-bound pools. Headline: strict-pool top-5 100 % against CERF UFE; admitting partly-documented crises (18 → 57) costs one country at top-5.

7 Results

Table 4 lists the ten countries ranked most overlooked in 2025 under the widest evidence-bound pool. Eight also appear on OCHA's CERF UFE 2024–25 selection. The two partly-documented countries (Zimbabwe, Malawi; $c = 0.67$) exhibit high rank-IQR — both sit in the *contested sector-starved* cell, where a single-number tool fails loudest because the answer depends sharply on which donor profile is applied.

#	Country	ISO3	Median rank	IQR	c
1	Sudan	SDN	2.5	3.5	1.00
2	Honduras	HND	2.5	3.3	1.00
3	Afghanistan	AFG	5.0	1.0	1.00
4	Zimbabwe	ZWE	5.0	8.8	0.67
5	Somalia	SOM	5.5	4.0	1.00
6	Haiti	HTI	6.0	2.5	1.00
7	Malawi	MWI	7.5	12.0	0.67
8	Mozambique	MOZ	9.0	4.5	1.00
9	South Sudan	SSD	9.0	1.8	1.00
10	Chad	TCD	9.5	7.8	1.00

Table 4: Top-10 most overlooked, 2025 (widest pool, $n = 57$). Low IQR (AFG 1.0, SSD 1.8) indicates profile consensus; high IQR (MWI 12.0, ZWE 8.8) indicates contested ranking.

8 Limitations

Stated in full, as a first-class part of the method.

- **Incompleteness is accepted, not imputed.** The completeness flag is attached to every score; partly-documented countries are scored alongside full-data ones under evidence-bounded scoring, and the trade-off is systematic under-ranking rather than inflation.
- **Cluster taxonomy drifts** between HNO and FTS. A simple normalisation (lower-case + non-alphanumeric → underscore) is used; a curated cluster map is the next step.
- **INFORM methodology rescaling**, February 2026, makes severity indices pre- and post-boundary not comparable. The ontology declares this as a failure mode; temporal aggregates that span the boundary use the stable ordinal category, not the continuous severity score.
- **Small sample.** The pool of country-years with the full six-attribute vector is on the order of tens. Statistical claims are made only against held-out external benchmarks.
- **Two significant figures.** A score of 0.62 is not meaningfully distinct from 0.59; median ranks are always reported with the IQR.
- **Not automation.** The system ranks; humans decide. Every displayed value carries its provenance.

9 Planned Next Step: A Learned Layer

The typed semantic layer is a feature store. The next step is a Level-6 family of *learned* composites above it, registered in the specification alongside the handcrafted ones. Three candidate tasks, ranked by experimental cleanliness:

1. **Structural recoverability.** Predict $G_p(u)$ from L1–L3 alone. Reproducible label; tests whether the MAUT formulation is learnable from observations. *Cleanest experiment.*
2. **Typology classification.** Four-class target from L1–L3. Same data, discrete label, easier

error analysis.

3. **Learning external curation.** Binary classification against CERF UFE membership. Most operationally interesting but noisiest, because the label is itself a political judgement.

Model family is constrained to regularised linear and gradient-boosted trees (small pool, interpretability-first). Temporal cross-validation (train $\leq$ 2024, test = 2025) and country-holdout are both used. The learned layer is never a replacement for the handcrafted scoring in the default view — both live side by side, with the learned model exposed as one of the lenses. A predicted ranking of overlookedness can, if communicated carelessly, reinforce existing patterns of neglect with an algorithmic gloss; the learned layer therefore ships with the same completeness-flag discipline as the rest of the ontology and a written use-and-limits note on any export.

10 Conclusion

When almost every active response is underfunded, a single coverage ratio stops discriminating. Recasting overlookedness as a joint distribution — across donor-preference profiles and across sector inequality within a crisis — surfaces the three kinds of overlooked crisis that scalar methods conflate, without requiring data the sector does not already publish. Implemented over a typed semantic layer so every displayed value traces to source data. Evidence-bounded so incomplete data cannot inflate rank. Eighty per cent top-5 overlap with OCHA's CERF UFE on the widest pool, 100 % on strict. What remains to be built is a learned layer above the ontology, held to the same interpretability and provenance discipline.

References

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